

Pricing Multi-Dimension Bermudan Options

LSM-FNN vs LSM-Poly

Brennan Davenport¹ and Nathaniel Levy¹

¹University of Georgia, USA , {Brennan.Davenport@uga.edu, Nathaniel.Levy@uga.edu}

December 16, 2025

Abstract

We study ... Briefly state the problem, your approach, and the main results.

1 Introduction

Explain the problem, why it matters, and what your contribution is. A common structure:

- What problem are you solving?
- Why is it hard / interesting?
- What do you contribute?

2 Related Work

Discuss prior approaches and how your work differs. Cite papers using [?].

3 Mathematical Setup

Let us first introduce a few definitions and assumptions. Our goal is to price multiple Bermudan options. For concreteness, we will refer to this collection as a basket of Bermudan options, or simply a basket option.

To make the model well-defined, we assume that the risk-free interest rate satisfies $r \geq 0$. Next, we define a set of exercise times

$$\mathcal{T} \subset [0, T],$$

which represents the dates at which the Bermudan option may be exercised.

We now define the payoff function. Let

$$h(S_t) : \mathbb{R}_+^d \rightarrow \mathbb{R}$$

denote the payoff function, where $S_t \in \mathbb{R}_+^d$ is the vector of underlying asset prices at time t . Furthermore, let $d \in \mathbb{Z}_{\geq 1}$ be the number dimension of this basket. The form of h depends on whether the option is a put or a call. In this project, we assume that the payoff is based on the average of the underlying assets.

- PUT option:

$$h(S_t) := \max \left(K - \frac{1}{d} \sum_{j=1}^d S_t^{(j)}, 0 \right)$$

- CALL option:

$$h(S_t) := \max \left(\frac{1}{d} \sum_{j=1}^d S_t^{(j)} - K, 0 \right)$$

Here, $K \in \mathbb{R}_+$ is the strike price.

With the payoff function defined, we can now define the value function for the basket option.

Let

$$V(t, S_t) : \mathcal{T} \times \mathbb{R}_+^d \rightarrow \mathbb{R}$$

denote the value function of a d -dimensional Bermudan option. The vector S_t represents the asset prices at time t .

Now that we have defined the value function V , we describe its form using the Bellman equation:

$$V(t_i, S_{t_i}) = \max \left(h(S_{t_i}), e^{-r(t_{i+1}-t_i)} \mathbb{E}[V(t_{i+1}, S_{t_{i+1}}) \mid S_{t_i}] \right),$$

where $t_i \in \mathcal{T}$. This provides an inductive definition of V . And we know that at the final exercise date $m = \max(\mathcal{T})$, we have

$$V(m, S_m) = h(S_m),$$

since there is no continuation beyond the last exercise date.

We now focus on the second term of the Bellman equation:

$$C_i(S_{t_i}) := e^{-r(t_{i+1}-t_i)} \mathbb{E}[V(t_{i+1}, S_{t_{i+1}}) \mid S_{t_i}].$$

The factor $e^{-r(t_{i+1}-t_i)}$ represents the discounted cash flow, which is straightforward (assuming a risk-free model). The challenging part is the conditional expectation

$$\mathbb{E}[V(t_{i+1}, S_{t_{i+1}}) \mid S_{t_i}],$$

which we refer to as the continuation value. In high dimensions, it is infeasible to compute this exactly using a lookup table.

In the classical Longstaff-Schwartz method, this conditional expectation is approximated using polynomial regression:

$$\mathbb{E}[V(t_{i+1}, S_{t_{i+1}}) \mid S_{t_i}] \approx \sum_{\alpha} \beta_{\alpha} \phi_{\alpha}(S_{t_i}),$$

where $\{\phi_{\alpha}\}$ is a set of polynomial basis functions of the current asset prices, and β_{α} are the regression coefficients.

The goal of this research is to replace the polynomial regression approximation with a feedforward neural network (FNN):

$$\mathbb{E}[V(t_{i+1}, S_{t_{i+1}}) \mid S_{t_i}] \approx \text{FNN}(S_{t_i}; \theta),$$

where θ represents the trainable parameters of the neural network. This approach allows us to efficiently approximate the continuation value even in high-dimensional settings.

4 Methodology

Describe your algorithm / model.

4.1 Model

Explain the model intuition first, then equations.

4.2 Algorithm

Pseudo-code or step-by-step explanation.

5 Experiments / Results

5.1 Experimental Setup

Data, parameters, assumptions.

5.2 Results

Tables, plots, and discussion.

Method	Price	Runtime
LSM (Poly)	10.42	0.31s
LSM (FNN)	10.57	0.48s

Table 1: Pricing results

6 Discussion

Interpret results, tradeoffs, limitations.

7 Conclusion

Summarize findings and suggest future work.